

High-Protein Penne with Ground Turkey & Mushrooms

This ground turkey pasta with mushrooms is a delicious, protein-packed meal, thanks to its powerhouse ingredients. Lean ground turkey teams up with cottage cheese, which not only boosts protein but also adds creaminess without relying on heavy cream. Mushrooms bring a savory depth of flavor while being low in calories, and whole-grain penne contributes fiber for a hearty and balanced meal!

By **Julia Levy** | Updated on September 6, 2025

 Reviewed by Dietitian **Emily Lachtrupp, M.S., RD**

Active Time: 30 mins

Total Time: 30 mins

Servings: 4

Nutrition Profile:

No Added Sugar Mediterranean Diet Sesame-Free Nut-Free Healthy Pregnancy Soy-Free High-Protein Egg-Free

Ingredients

- 2 pints cherry tomatoes
- 4 cloves garlic, peeled and smashed
- 1 teaspoon Italian seasoning, divided
- 4 tablespoons extra-virgin olive oil, divided
- 8 ounces whole-wheat penne
- 1 pound ground turkey
- $\frac{1}{8}$ teaspoon salt plus $\frac{1}{4}$ teaspoon, divided
- 1 (8-ounce) package sliced cremini mushrooms
- $\frac{1}{2}$ teaspoon fennel seed
- 1 cup full-fat cottage cheese
- $\frac{1}{2}$ cup grated Parmesan cheese
- Fresh oregano for garnish (optional)

Directions

Step 1

Preheat oven to 425°F. Toss 2 pints cherry tomatoes, the smashed garlic and ½ teaspoon Italian seasoning with 1½ tablespoons oil on a large rimmed baking sheet. Roast until the tomatoes burst and are juicy and lightly caramelized, 18 to 20 minutes.

Step 2

Meanwhile, bring a medium saucepan of water to a boil. Add 8 ounces penne; cook according to package directions. Drain, reserving ½ cup cooking water.

Step 3

Heat 1 tablespoon oil in a large skillet over medium-high heat. Add 1 pound ground turkey and ⅛ teaspoon salt; cook, stirring once or twice, until browned and no longer pink, about 5 minutes. Transfer to a bowl; do not wipe the skillet clean.

Step 4

Add the remaining 1½ tablespoons oil and 1 (8-ounce) package mushrooms to the skillet; arrange in a single layer and cook, undisturbed, until browned on the bottom, about 3 minutes. Stir in ½ teaspoon fennel seed and the remaining ½ teaspoon Italian seasoning; continue to cook, stirring occasionally, until browned all over, about 3 more minutes. Remove from heat and stir in the browned turkey.

Step 5

Combine 1 cup cottage cheese and ½ cup Parmesan in a mini food processor; process until smooth, 15 to 20 seconds.

Step 6

Scrape the roasted tomatoes, garlic and any accumulated juices from the pan into the skillet with the turkey mixture. Stir everything together, scraping up any browned bits from the bottom of the skillet. Add ½ cup cottage cheese mixture, ¼ cup reserved cooking water and the remaining ¼ teaspoon salt; stir until fully incorporated. Fold in the pasta; cook over medium-high heat, stirring often and adding the remaining ¼ cup reserved cooking water, 1 tablespoon at a time, as needed, until the pasta is evenly coated and the sauce clings to the pasta, 1 to 2 minutes.

Step 7

Divide the pasta among 4 bowls and dollop with the remaining cottage cheese mixture. Garnish with oregano, if desired.

Frequently Asked Questions

Do I have to make this recipe with turkey?



We love the flavor that turkey provides in this recipe, but you can try substituting it with ground chicken, pork or beef.

How should I store and reheat ground turkey pasta leftovers?



Ensure the leftovers are cooled down before placing them in an airtight container and refrigerating. For the best results, consume the leftovers sooner rather than later, although it's

generally safe to refrigerate leftovers for up to 4 days. When ready to reheat, they can be warmed in a skillet over medium heat. You should add a little water, too.

What should I serve with ground turkey pasta?



This dish rounds out well with sheet-pan vegetable recipes. It's also excellent with Balsamic Oven-Roasted Carrots or one of our favorite salads.

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Nutrition Facts

Per serving: **Serving Size 2 cups pasta mixture & 2 Tbsp. cottage cheese mixture** 624 calories; total fat 25g; saturated fat 6g; cholesterol 100mg; sodium 709mg; total carbohydrate 49g; dietary fiber 7g; total sugars 7g; added sugars 0g; protein 56g; vitamin c 23mg; calcium 201mg; iron 4mg; potassium 1137mg
Additional reporting by Sarah Pflugrad, Ph.D., RDN, CSCS and Linda Frahm

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